Fiscal Year:	FY 2020	**Task Last Updated:**	FY 10/25/2019
PI Name:	Swan, James Ph.D.		
Project Title:	Dynamic Self-assembly Driven by Time with Varying Fields		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02139-4301	**Congressional District:**	7
Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2015-16 Physical Sciences NNH15ZTT001N-15PSI-C: Use of the NASA Physical Sciences Informatics System – Appendix C
Start Date:	12/01/2017	**End Date:**	11/30/2020
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/30/2020 per NSSC info (Ed., 11/12/19)		
Key Personnel Changes/Previous PI:	October 2019 report: The project entered its second year. Zachary Sherman completed his Ph.D. at MIT and a new second year graduate student, Emily Krucker began working on the proposed tasks. October 2018 report: The project began this year and a 4th year graduate student in chemical engineering at MIT, Zachary Sherman, started work on the proposed tasks.		
COI Name (Institution):			
Grant/Contract No.:	80NSSC18K0162		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This proposal is a research plan for using theory and simulation to analyze and extend the InSPACE 1, 2, 3, and 3+ investigations of the self-assembly of magnetically responsive nanoparticles in a time-varying magnetic field. The original work found that self-assembly was greatly accelerated by utilizing a simple open-loop control scheme of the magnetic interactions. This methodology offers one solution to the difficult engineering challenge of developing scalable and cost-effective fabrication methods for structured nanomaterials with proven applications in optoelectronics, energy storage and harvesting, biological sensing, 3D printing, and high-performance coatings. However, because of limited spatial and temporal resolution in the experiments, detailed analysis of the self-assembled structures was impossible. A deeper understanding of out-of-equilibrium self-assembly demonstrated in the InSPACE suite is needed before it can be leveraged to design processes that rapidly produce high quality nanomaterials. The proposed work has two main objectives. First, we will take advantage of insights developed since the InSPACE experiments for other out-of-equilibrium assemblies to re-evaluate the InSPACE data in the Physical Sciences Informatics (PSI) system. This will reveal new results that the original InSPACE team was unable to obtain. Second, we will use the InSPACE data in the PSI system to develop and validate a computational model of nanoparticle self-assembly in time-varying fields and use that model to extend the investigation of toggled magnetic field assembly beyond the parameter space explored in the experimental suite. This will aid in developing a predictive theoretical framework for dynamic self-assembly that can facilitate design of rapid nanomaterial syntheses. The work will utilize GPU-accelerated discrete element simulations of particle self-assembly driven by magnetic fields with detailed models of the magnetic and hydrodynamic interactions among the particles. The Principal Investigator's (PI) laboratory has recently developed new algorithms for modeling these interactions that have accelerated simulations by roughly a factor of 100 times. Consequently, discrete element simulations of the hierarchical self-assemblies observed in the InSPACE experiments are possible. The video micrographs from InSPACE are of limited spatial and temporal resolution; therefore, simulations capable of resolving the particle structure and dynamics on a local scale while replicating structure and growth kinetics on larger length scales and longer time scales provide additional insight into the InSPACE data. The ability to translate InSPACE microgravity data for the many-bodied, dynamic interactions among self-assembling particles into well-validated models for the same experiments will enable simulations to predict and then design dynamic self-assembly processes terrestrially as well as during continued space operations. The PI’s laboratory has already demonstrated a predictive framework for other types of toggled self-assembly and is currently developing theories for the InSPACE assembly that can be used to predict the outcome and quantify the robustness of dynamical self-assembly processes. Incorporation of microgravity data into this theory is essential for testing its predictive capability. The PI was part of the original InSPACE 2 team as a post-doctoral scholar at the University of Delaware before starting as an assistant professor at Massachusetts Institute of Technology (MIT). He is first author of the two published papers on InSPACE experiments and is intimately familiar with the project. He has also published several papers extending toggled self-assembly to ground experiments and to simulations of new systems. One MIT graduate student, Zachary Sherman, currently completing a Ph.D. focused on toggled self-assembly processes will assist in the research. Their extensive experience makes the investigators uniquely qualified to re-examine and extend the InSPACE analysis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed work focuses on far from equilibrium methods for assembling nano-particles. The goal is to discover routes to self assembly free of kinetic constraints introduced by equilibrium thermodynamics. If successful, these pathways will enable the production of microstructures materials with few defects at rates not accessible by current technological platforms. The potential implications for these materials technologies span advance manufacturing in the form of 3D printing, the development of high performance coatings, and the creation of functional or tunable photonics and phononic materials for micro-devices.
Task Progress:	The proposed work concerns the development of fundamental models for how small particles self assemble in fluids to form useful structures. The applications range from structural materials, engineered for strength, to optical materials, engineered for performance in optics applications. Small particles can be made to distribute themselves in a many different ways within a composite material and directly affect these basic properties the material. One mode by which particles can be made to self assemble into prescribed positions in a composite is through use of external, uniform fields. If the particles are magnetizable, then magnetic fields do this positioning. The physical nature of the self-assembled structures have been difficult to determine through experimentation alone. Specialized experiments in space have been required in the past to eliminate terrestrial effects such as gravity and magnetic interference. We used data from these experiments to validate new simulations and theories developed these last few years. One circumstance we examined is how spherical, paramagnetic particles self assemble in a steady, uniform magnetic field. We found that a principle called mutual polarization has a crucial effect on the strength of the magnetic field required to drive this sort of self assembly. We developed a new model of the thermodynamics of these materials that predicts the field strength for self assembly quantitatively. Additionally, we tested the model against computer simulations that were in turn validated against data collected in NASA experiments. Another circumstance we examined is how particles self assemble in an unsteady, uniform magnetic field. We performed numerous computer simulations meant to replicate and extend experiments from the NASA InSpace 1 and 2 missions. These simulations were used to show that new, and previously unknown, states of matter in magnetic particle dispersions can be realized with time varying control over the external field. A final problem we studied directly using these simulation tools is the effect of a rotating magnetic field on dispersions of magnetic colloids. These simulations were inspired by InSpace 3 data in which large aggregates of magnetic colloids were driven to reorient by Earth's magnetic field as the space station orbited. We examined a slightly different case where in a uniform magnetic field was made to rotate at a controlled rate and we measured the rate of transport of particles in the dispersion. We found that transport was maximized by the changing field direction over a precise range of rotation rates, and we developed an analytical model to predict this optimum point in terms of the properties of the particle and the magnetic field. This equation is now being used by researchers doing terrestrial experiments in which such rotating fields are used to deliver cargo (small molecule pharmaceuticals) through porous tissues to regions of infection.

	Finally, we extended the simulation tools developed and validated as part of this project to a broader range of physical circumstances. Namely, we enabled the simulation of non-spherical particles in magnetic fields and the simulation of polarizable particles in complex electrolytes. We are now conducting simulations to estimate the strength of the interactions between particles in fields and complex media like electrolytes, which we hope will lead to the development and design of better energy infrastructure such as batteries.
Bibliography Type:	Description: (Last Updated: 04/30/2025)
Articles in Peer-reviewed Journals	Sherman ZM, Pallone J, Erb RM, Swan JW. "Enhanced diffusion and magnetophoresis of paramagnetic colloidal particles in rotating magnetic fields." Soft Matter. 2019 Sep;15(33):6677-89. https://doi.org/10.1039/C9SM00890J , Sep-2019
Articles in Peer-reviewed Journals	Sherman ZM, Swan JW. "Transmutable colloidal crystals and active phase separation via dynamic, directed self-assembly with toggled external fields." ACS Nano. 2019 Jan;13(1):764–71. https://doi.org/10.1021/acsnano.8b08076 , Jan-2019